BINUS UNIVERSITY
In Partnership with ASO COLLEGE GROUP JAPAN
BINUS ASO SCHOOL OF ENGINEERING

STEREO VISION FOR

VEHICLE DETECTION TASK - COMPUTER VISION

MEMBERS

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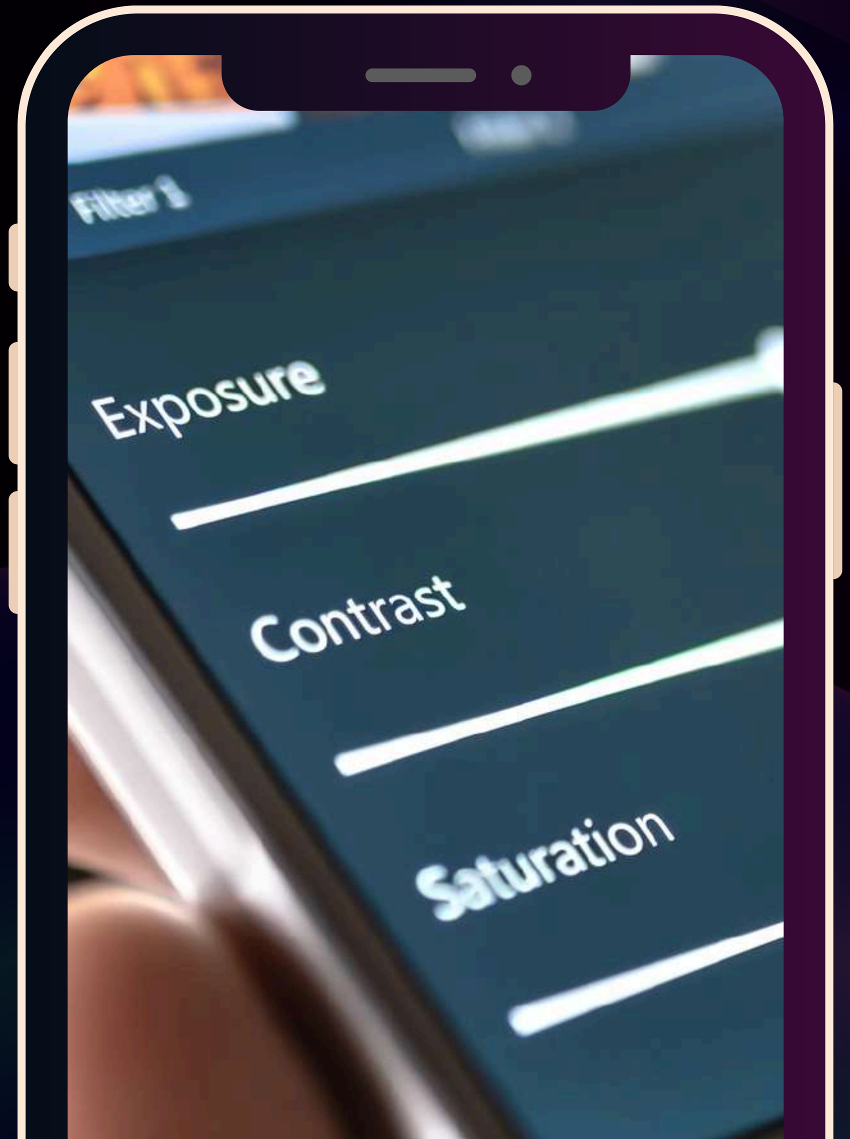
BACKGROUND

- Computer vision enables vehicles to perceive and understand the surrounding environment
- Vehicle detection is performed to identify cars and trucks in real time
- Stereo vision is used to estimate object distance by computing depth information
- Combining object detection with stereo depth allows distance measurement in meters
- Vision-based perception provides essential input for sensor fusion and vehicle control

VEHICLE DETECTION

- The system performs vehicle detection (car and truck) using a stereo vision setup with two identical webcams.
- Images are enhanced using manual feature extraction (contrast enhancement, edge emphasis).
- Manually designed convolutional filters highlight important vehicle features.
- YOLOv8n is used for training and detection due to its fast and lightweight architecture.

IMAGE ENHANCEMENT



Grayscale Conversion

- Simplifies processing and reduces computation
- Focus on Intensity Patterns

Histogram Equalization

- Improves contrast in low-light or uneven lighting conditions
- Makes vehicle edges and textures more visible

Gaussian Blur

- Reduces high-frequency noise
- Prevents false edges before edge detection

FEATURE EXTRACTION

Sobel Edge Enhancement

- Highlights strong horizontal and vertical edges
- Emphasizes vehicle contours such as body edges, wheels, and outlines

Edge Normalization

- Scales edge strengths to a standard range
- Improves consistency across different images

Feature-Level Enhancement (HOG)

- Capturing gradient orientation distributions
- Representing vehicle shape and texture robustly
- Being invariant to small illumination changes

BLOCK DIAGRAM (CV)

Stereo Camera

↓

Preprocessing
(Rectification)

↓

Vehicle Detection
(Car & Truck)

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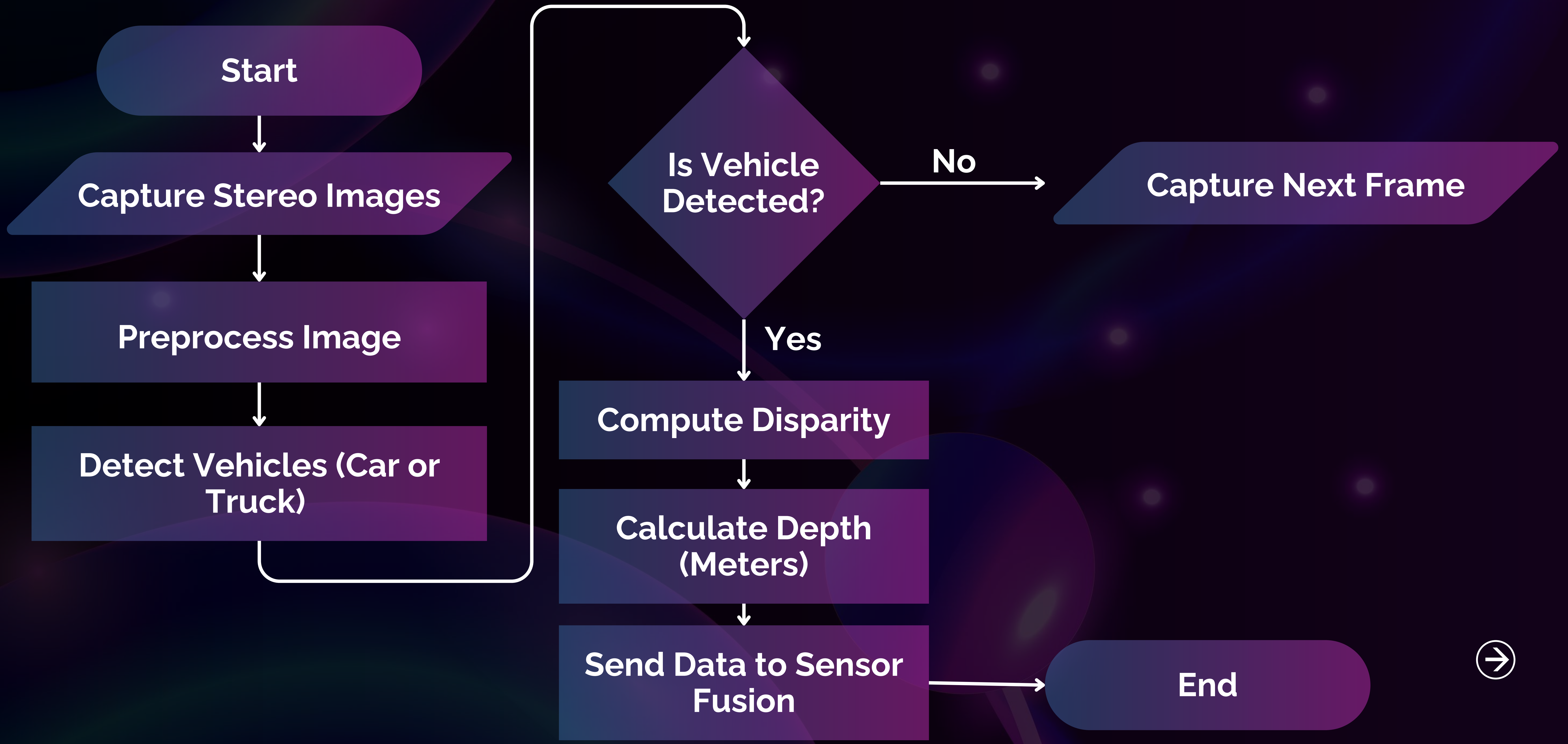
Stereo Matching
/ Disparity

↓

Depth (Meters)

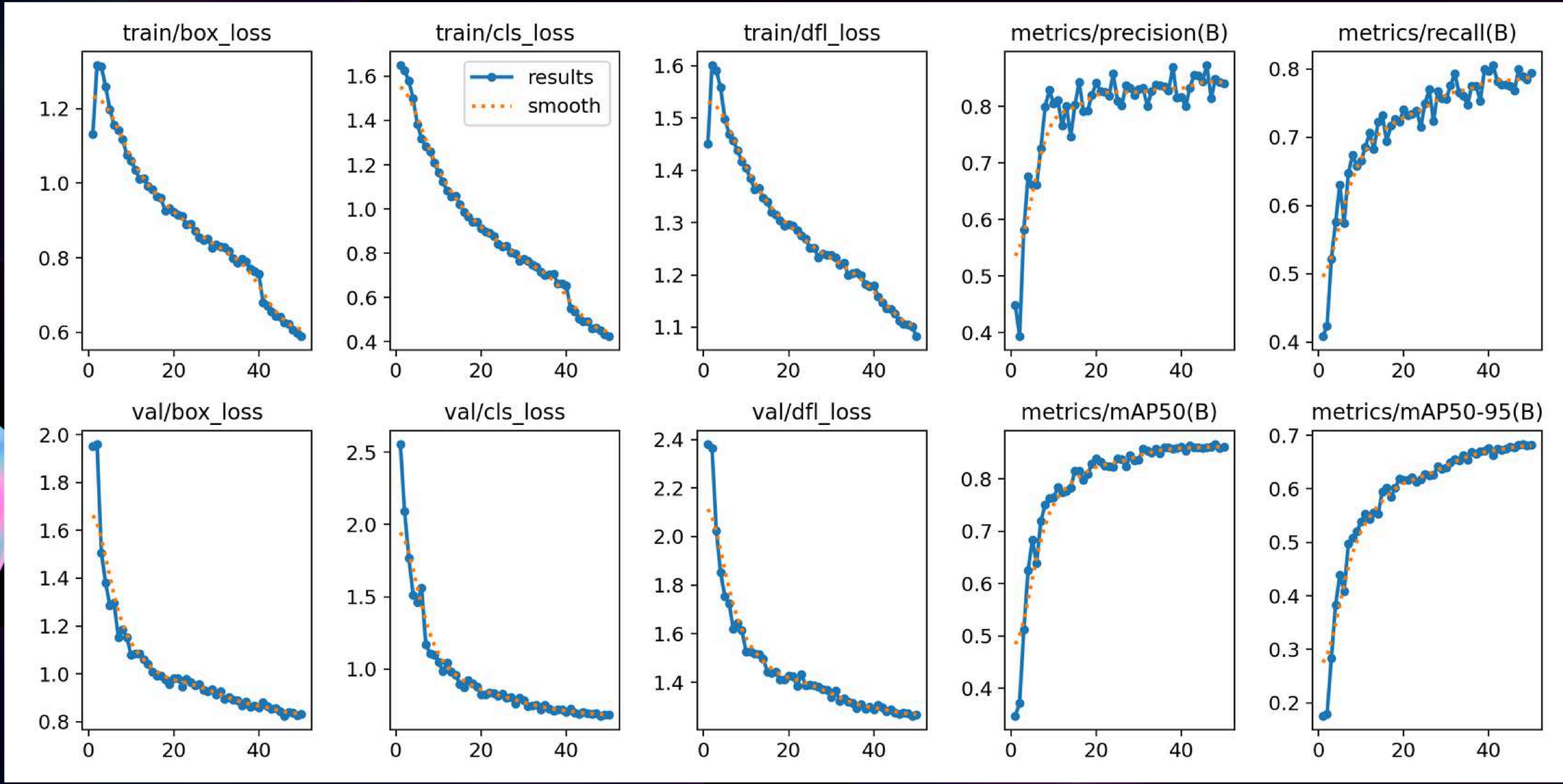
FLOWCHART (CV)



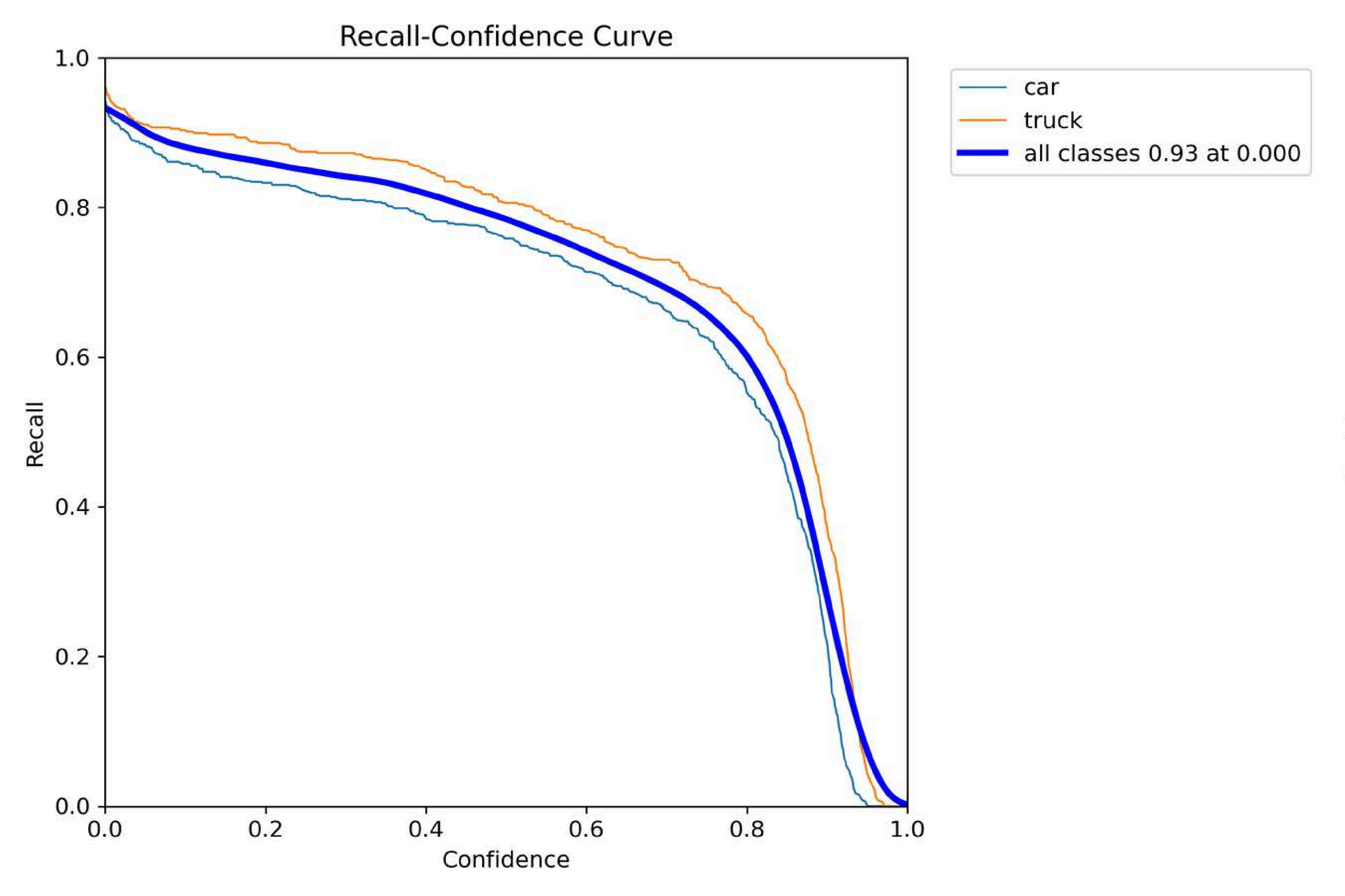
TRAINING

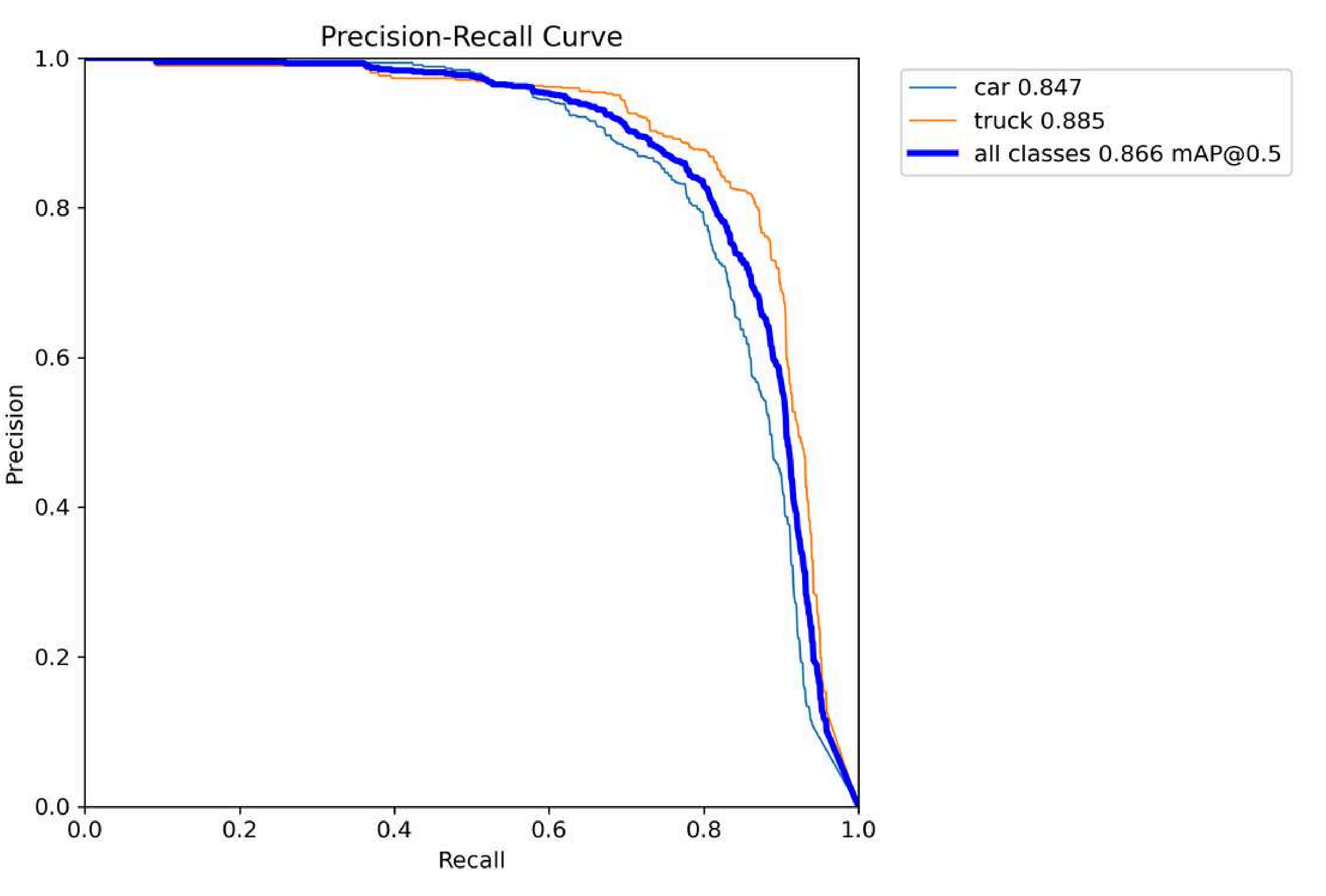
Train & Test	Epoch	Class
2531 - Training Image 361 - Test Image	50	0 - Car 1 - Truck

RESULT OF YOLO

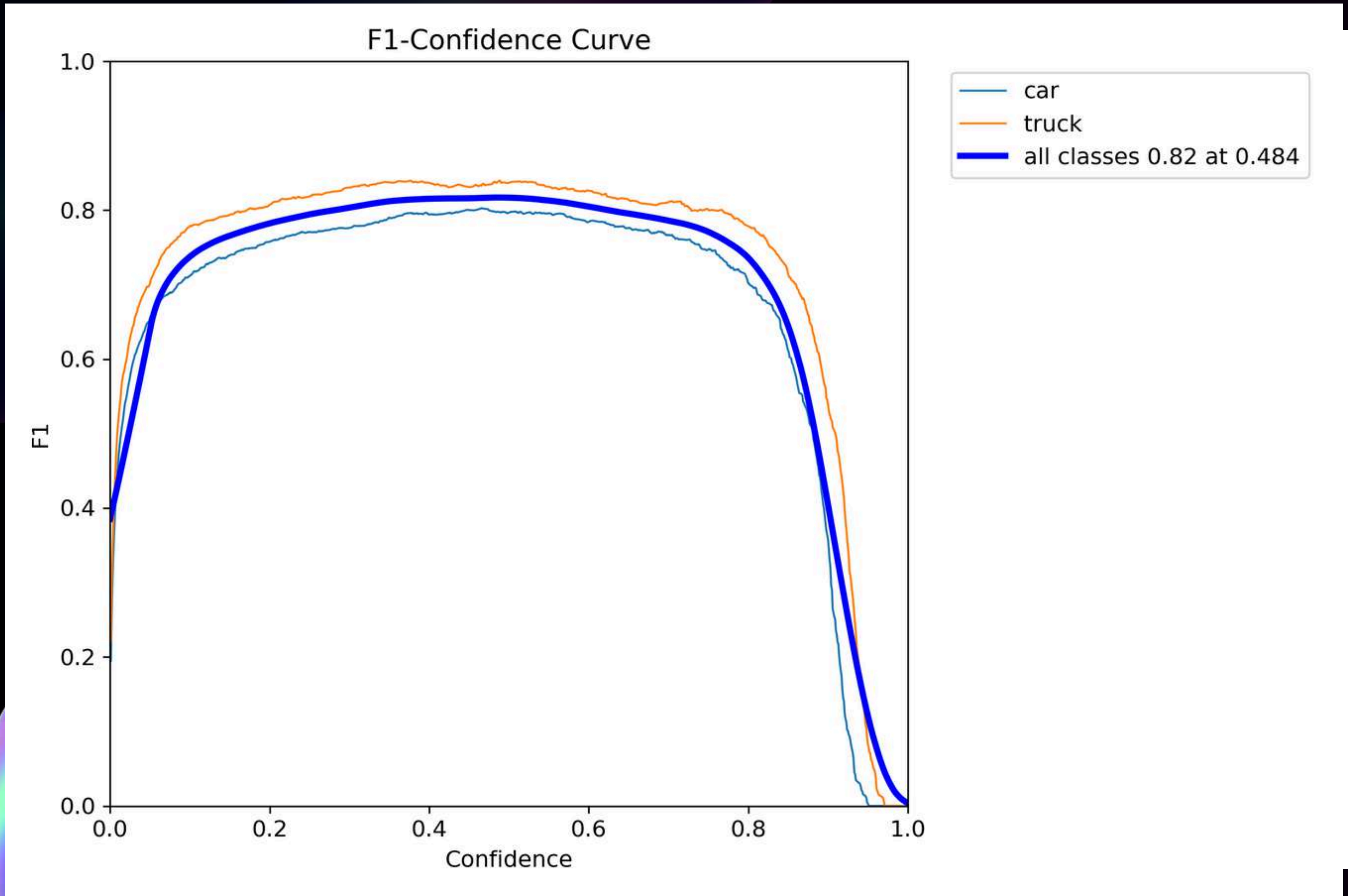


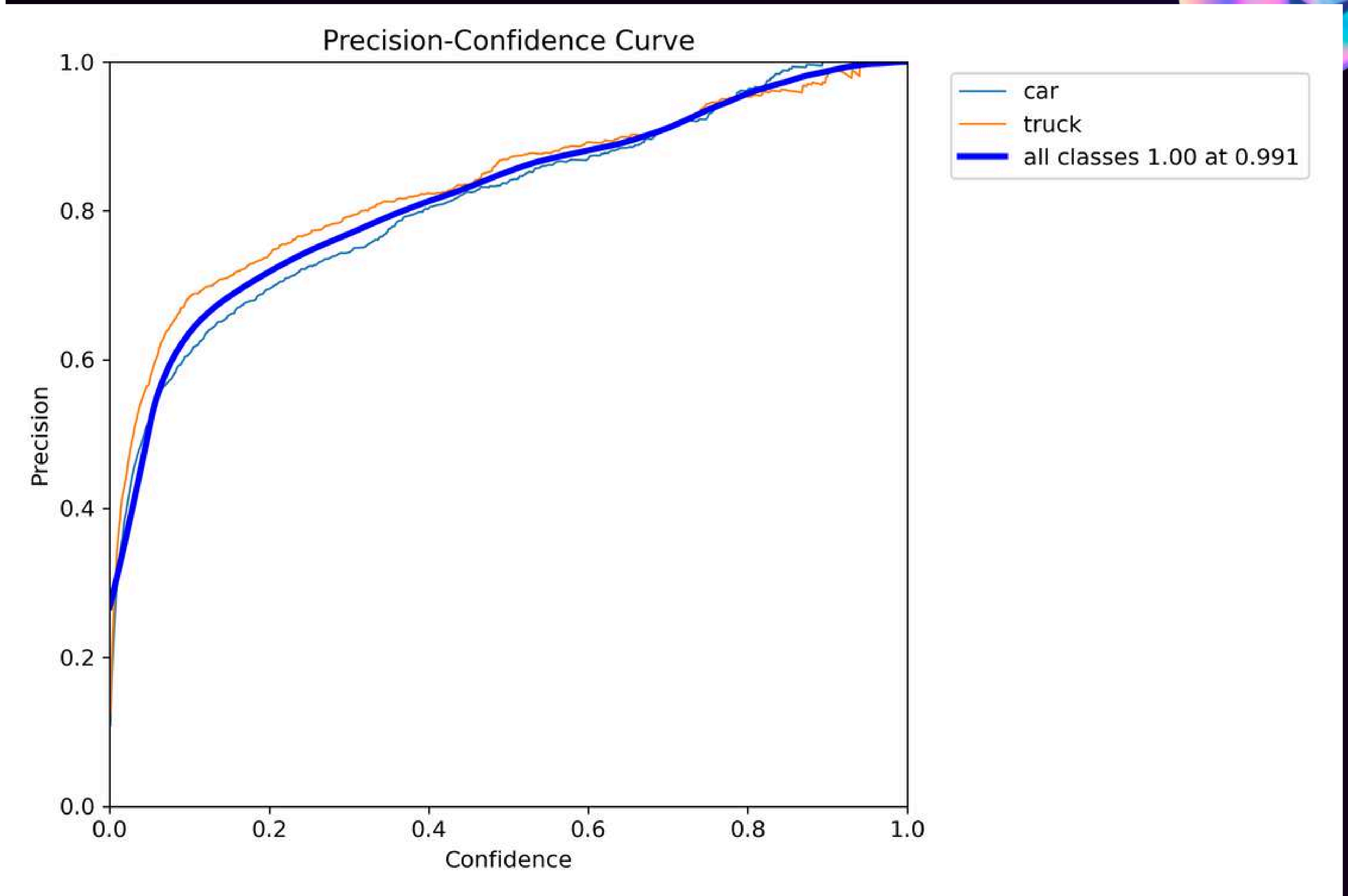
RESULT OF YOLO



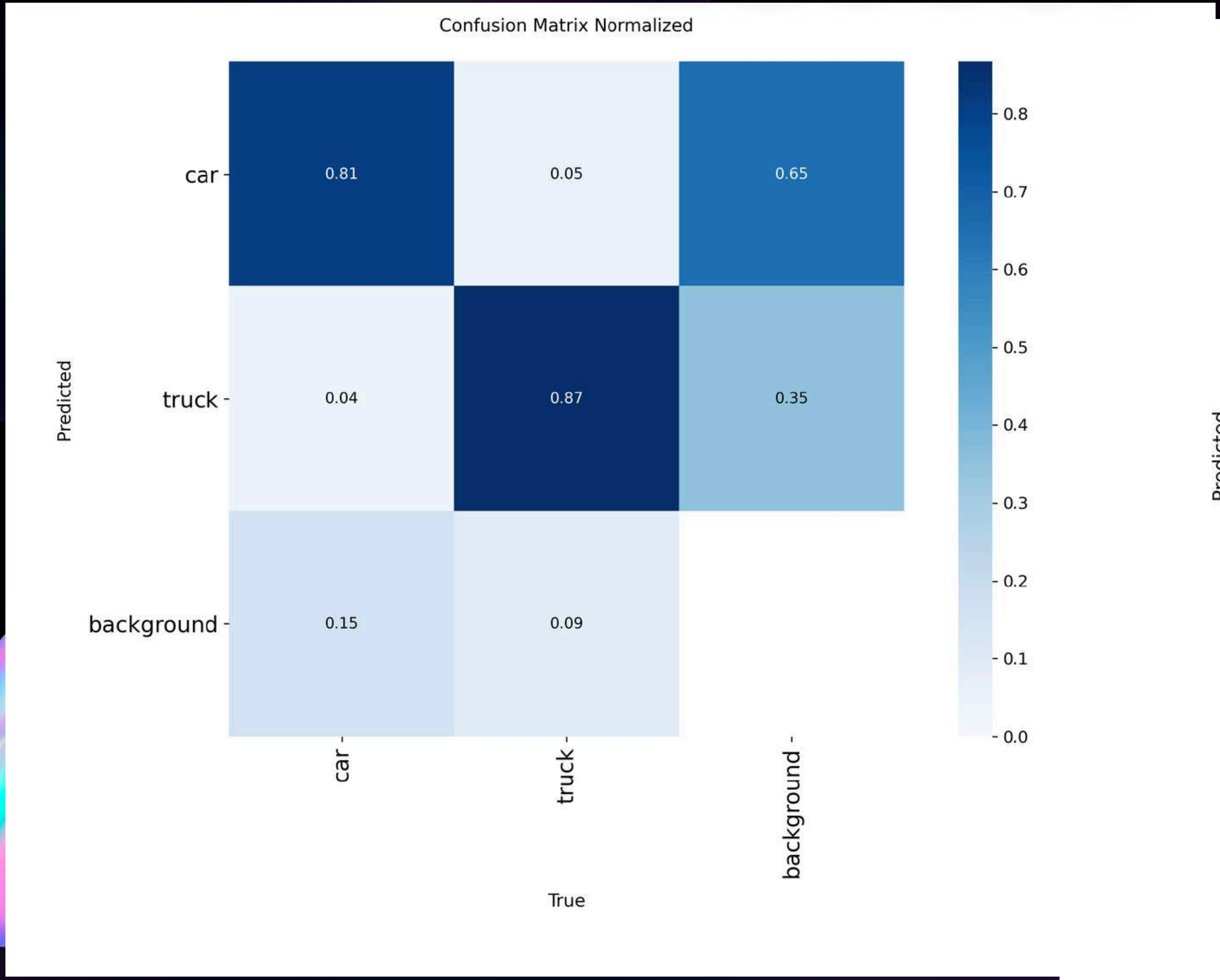


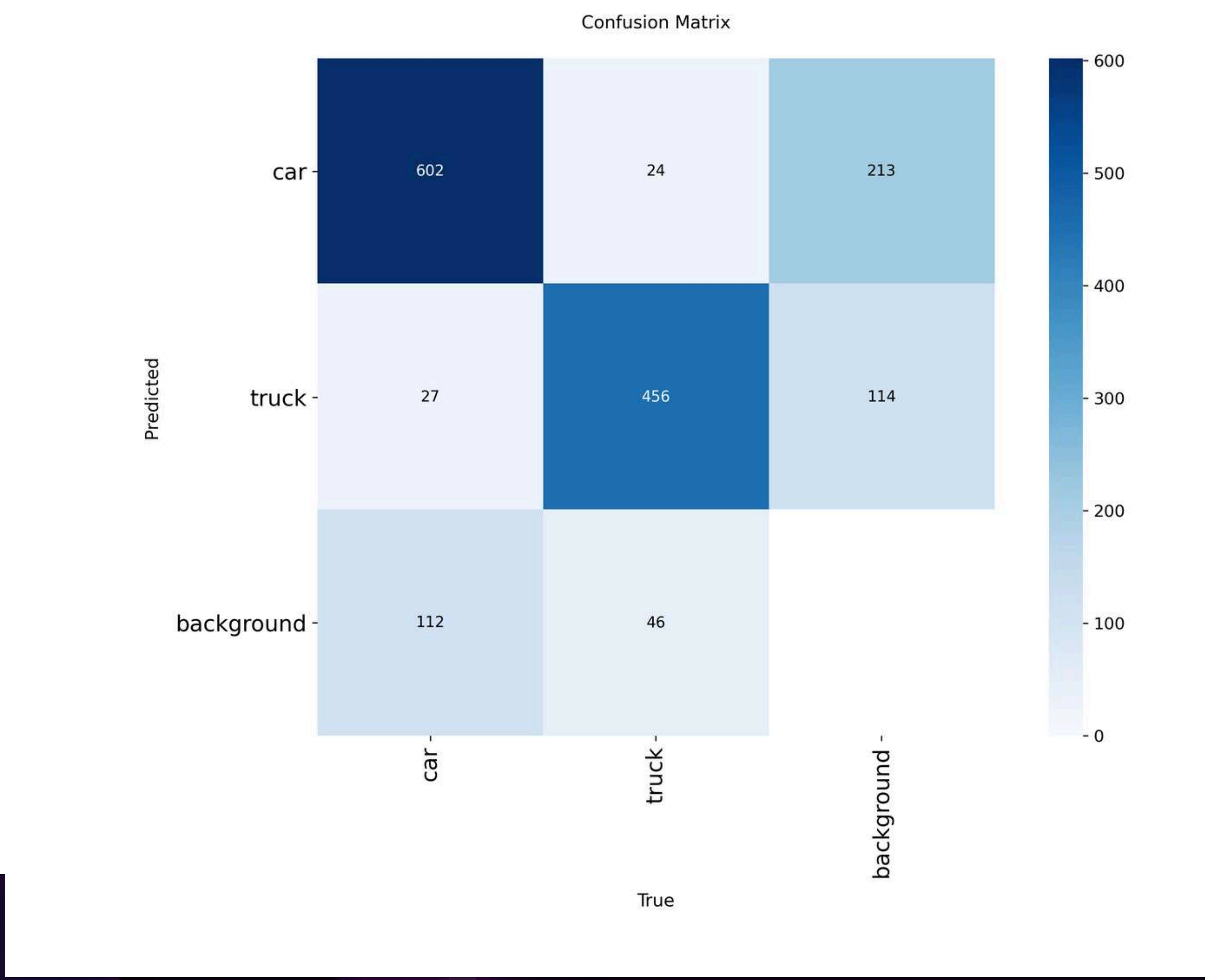
RESULT OF YOLO



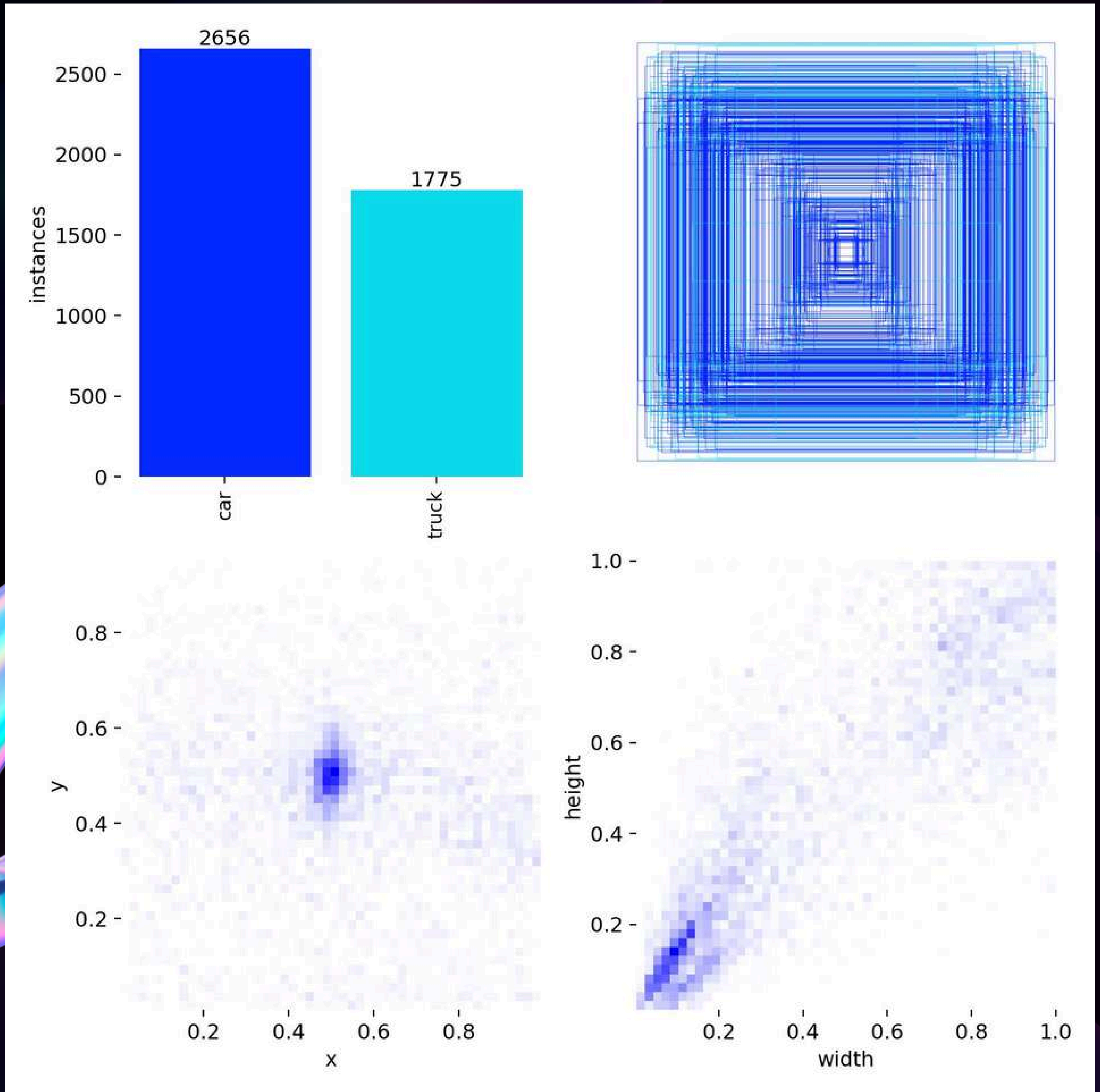


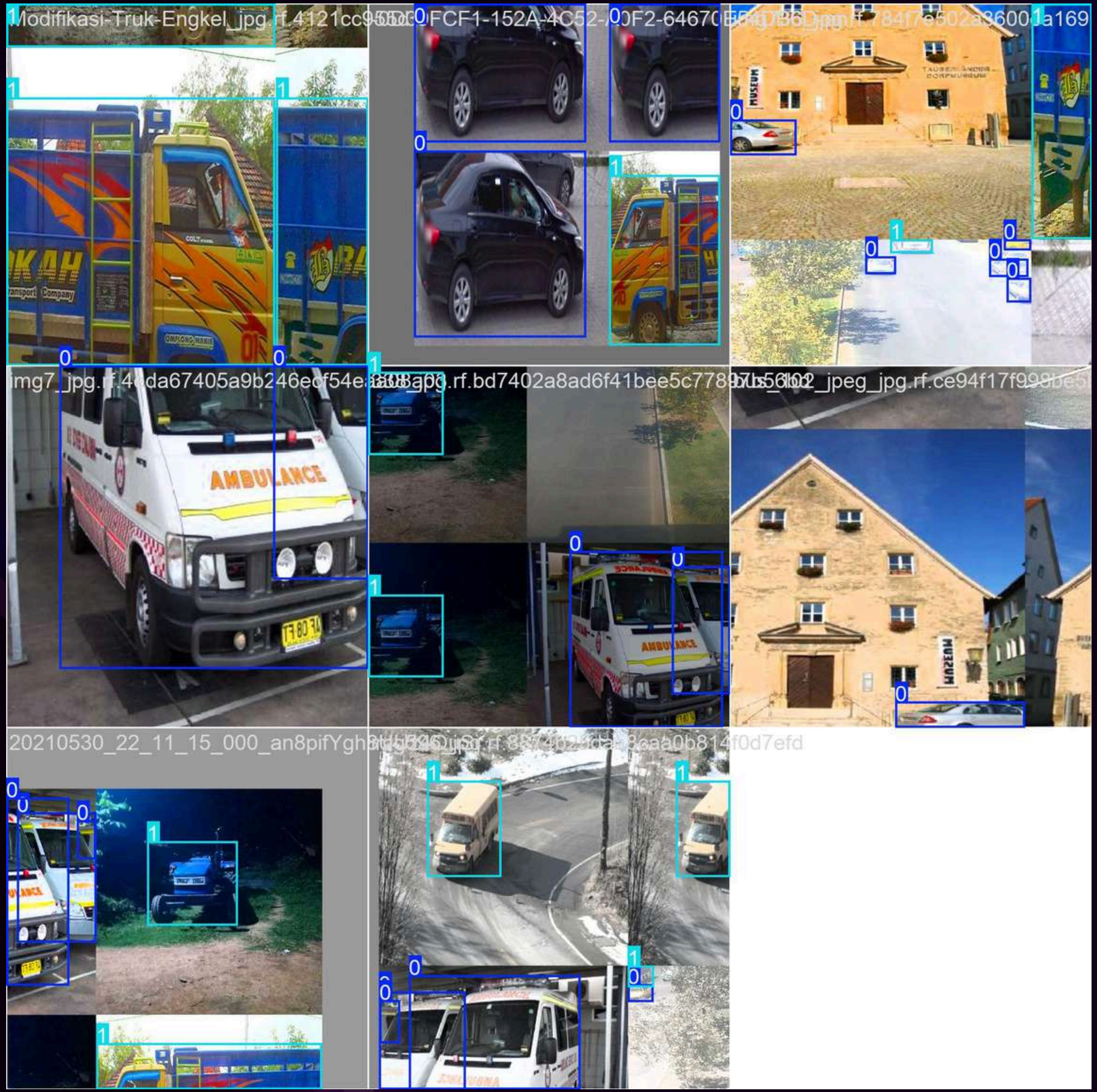
RESULT OF YOLO





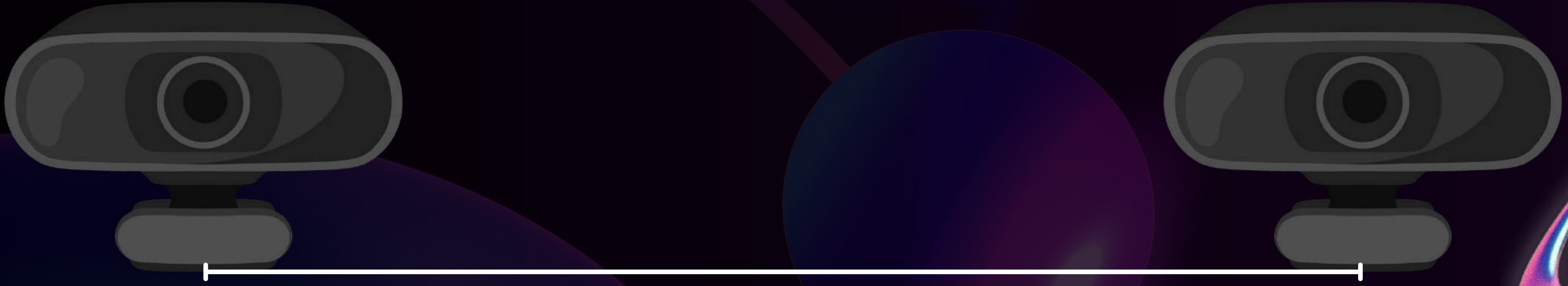
RESULT OF YOLO





STEREO VISION

- Two webcams with identical specifications capture synchronized left and right images.
- Stereo calibration and rectification align both views for accurate geometry.
- Stereo vision provides depth information, while YOLOv8n performs object detection.
- The combination enables robust vehicle detection with spatial awareness.



RESULTS

Stereo RMS

Stereo RMS: 0.9896 px
Euler Angles (deg)
Roll : 3.295
Pitch : -5.697
Yaw : -3.854

Mean reprojection error: 0.0422 px

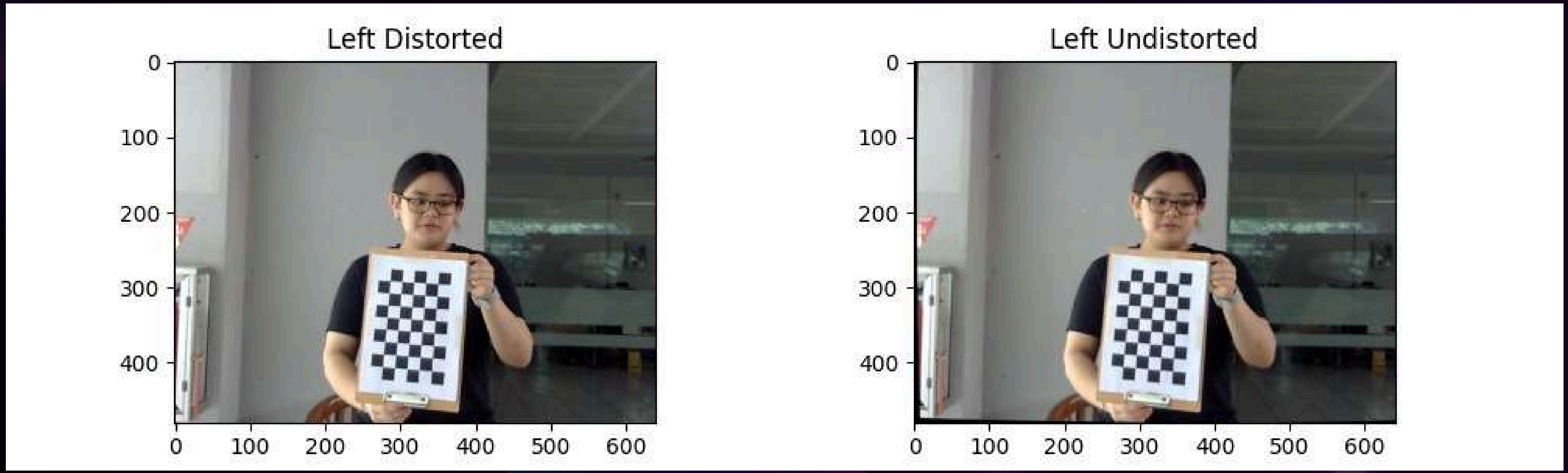
Individual Results

Calibrating mono cameras...
Left RMS : 0.2886
Right RMS: 0.3092

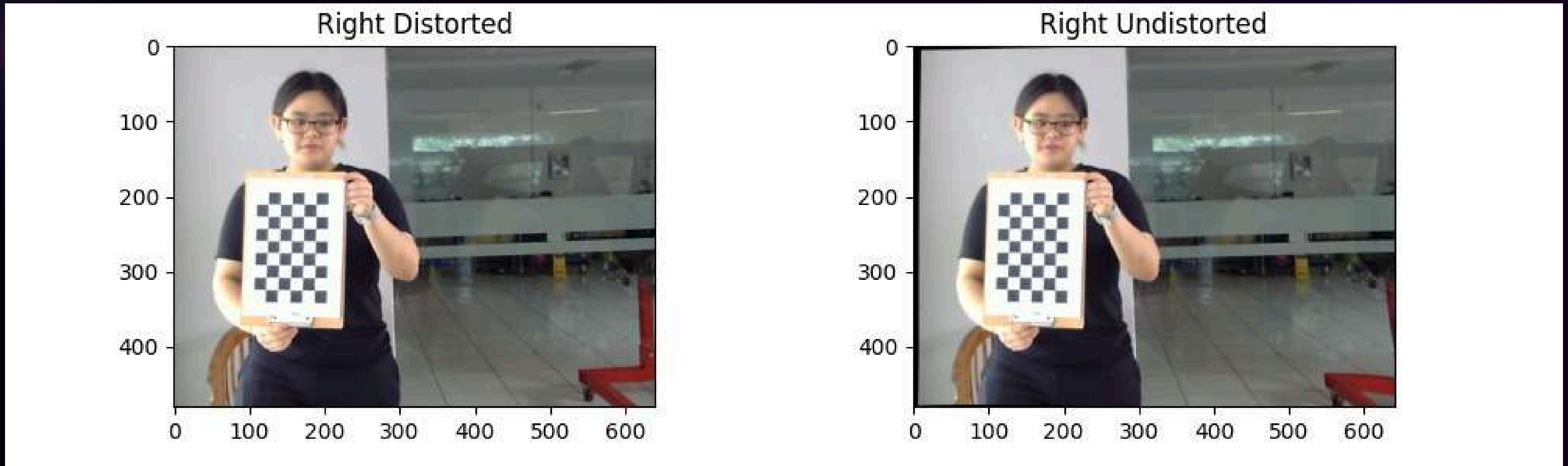
Baseline : 0.1035 m

Rotation Matrix R:
[[0.99862409 0.01955024 -0.04865922]
[-0.01871081 0.99966918 0.01764739]
[0.04898814 -0.01671265 0.99865953]]

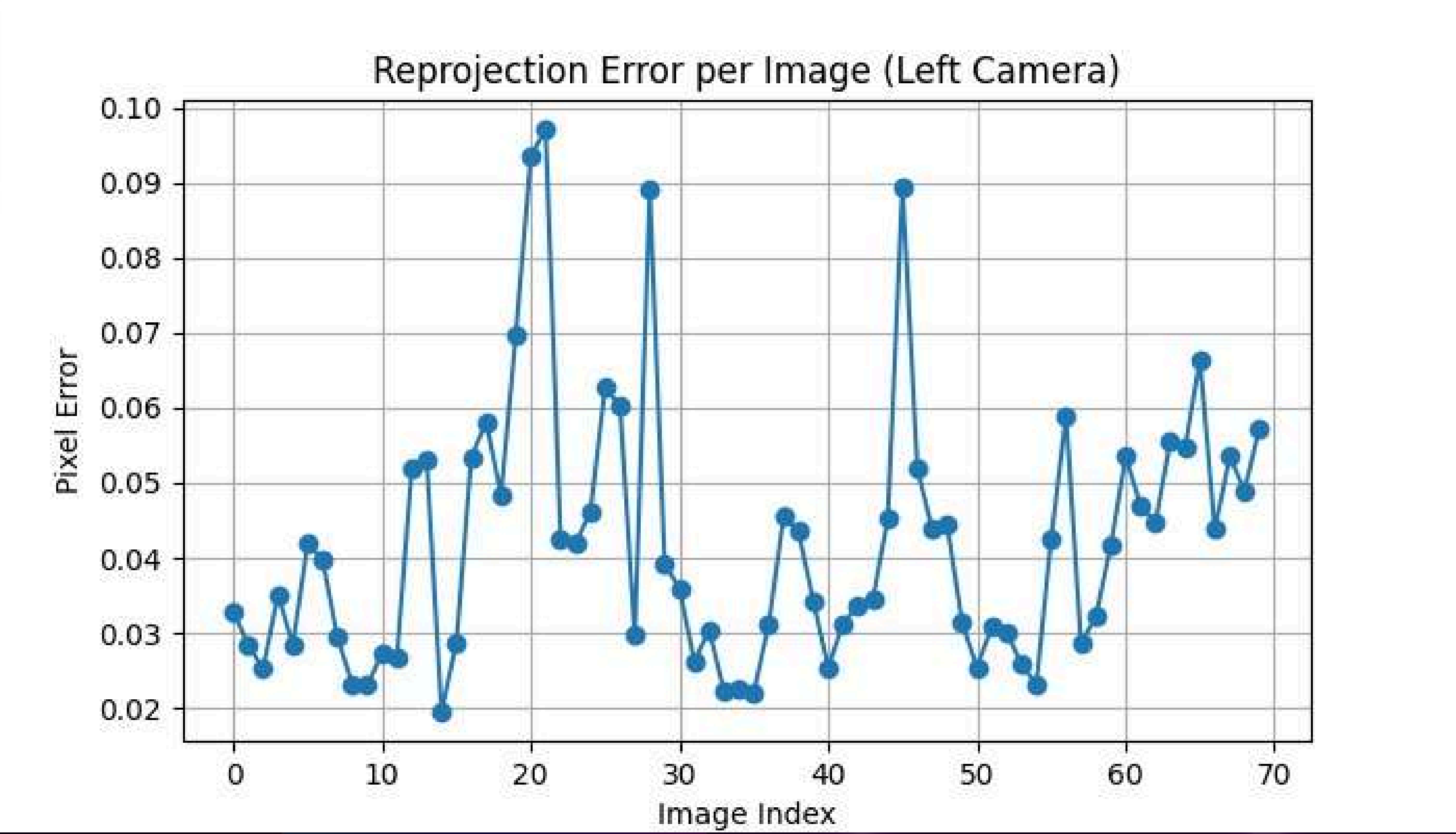
UNDISTORT IMAGE L



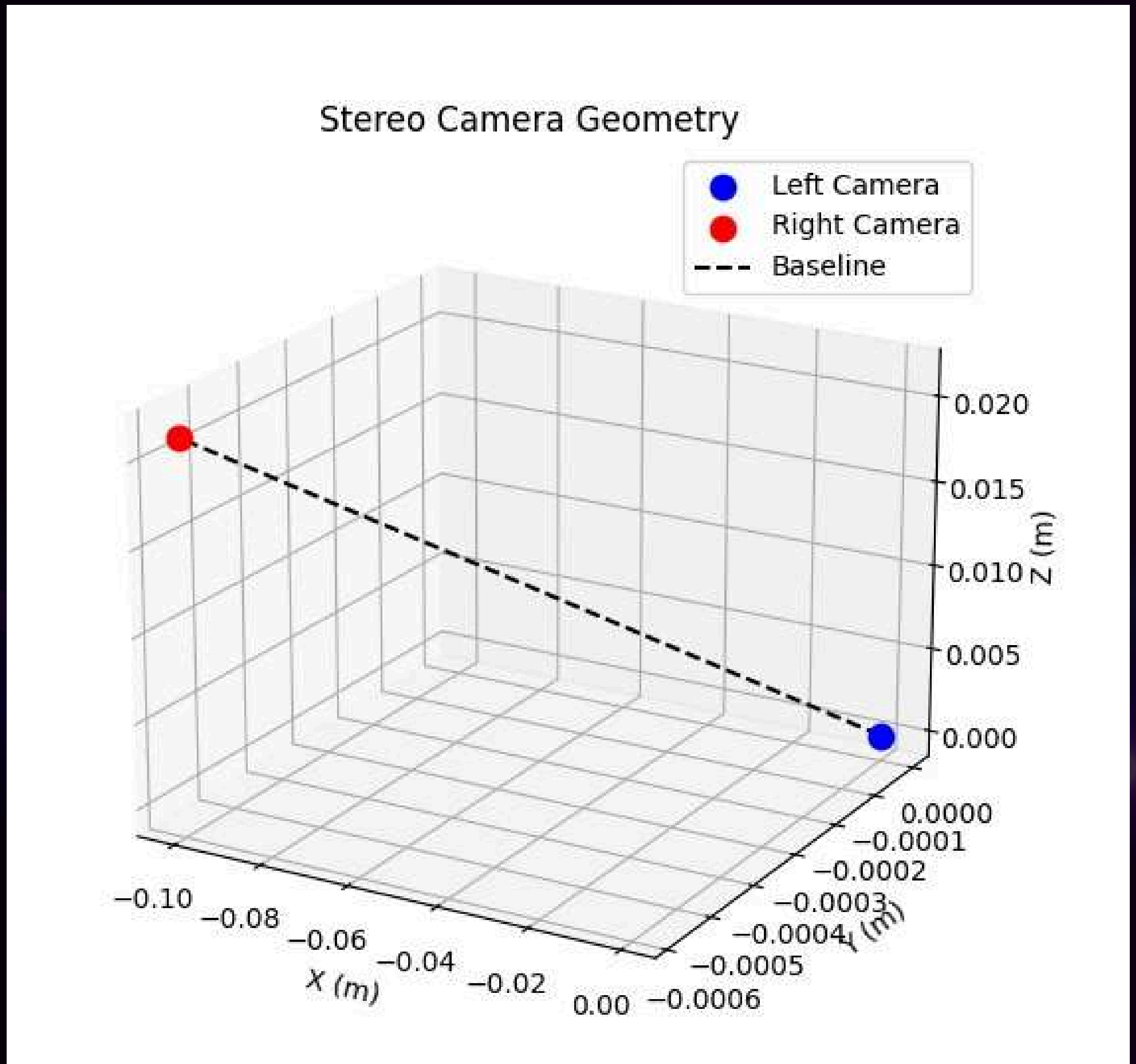
UNDISTORT IMAGE R



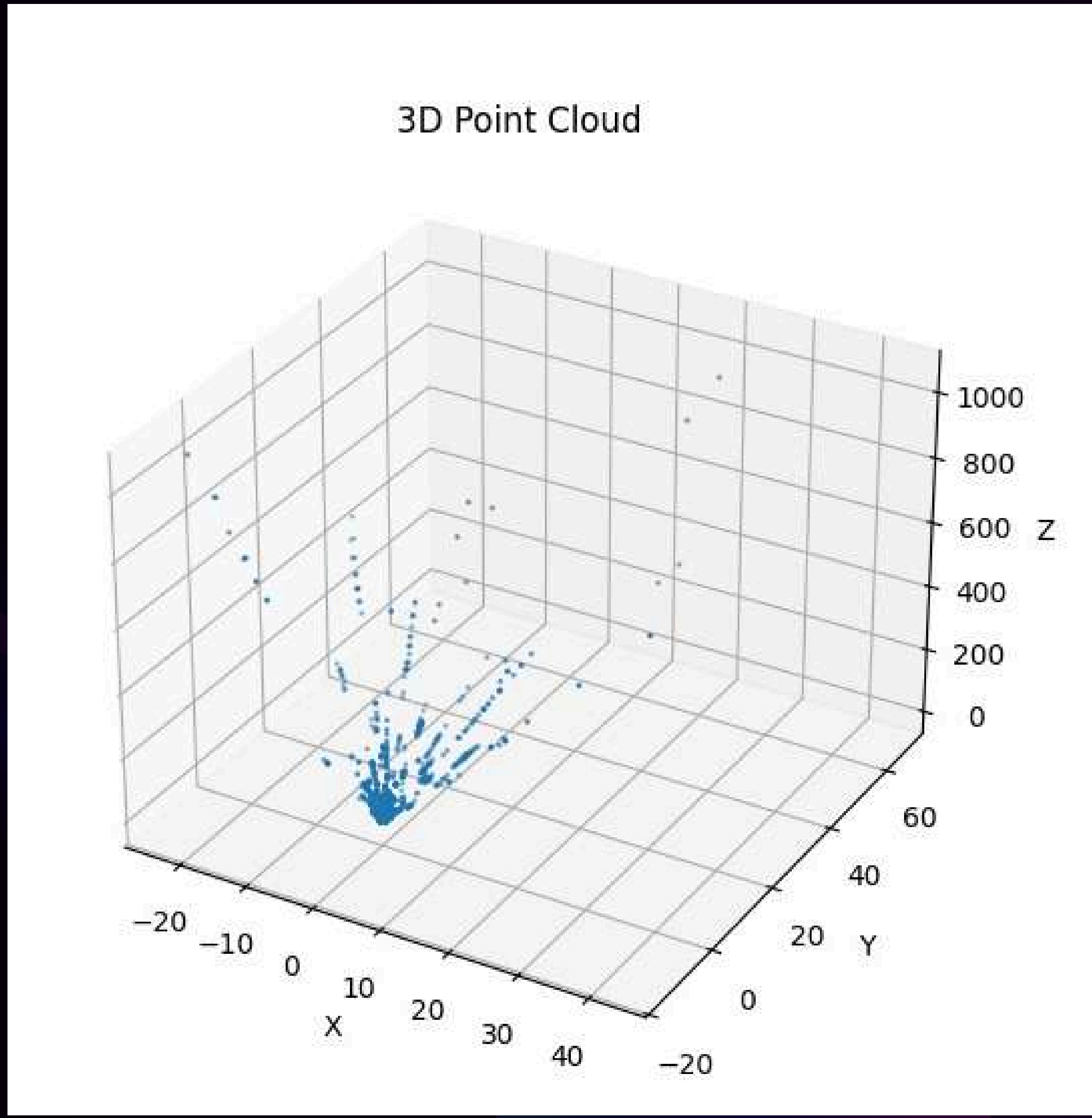
REPROJECTION ERROR L



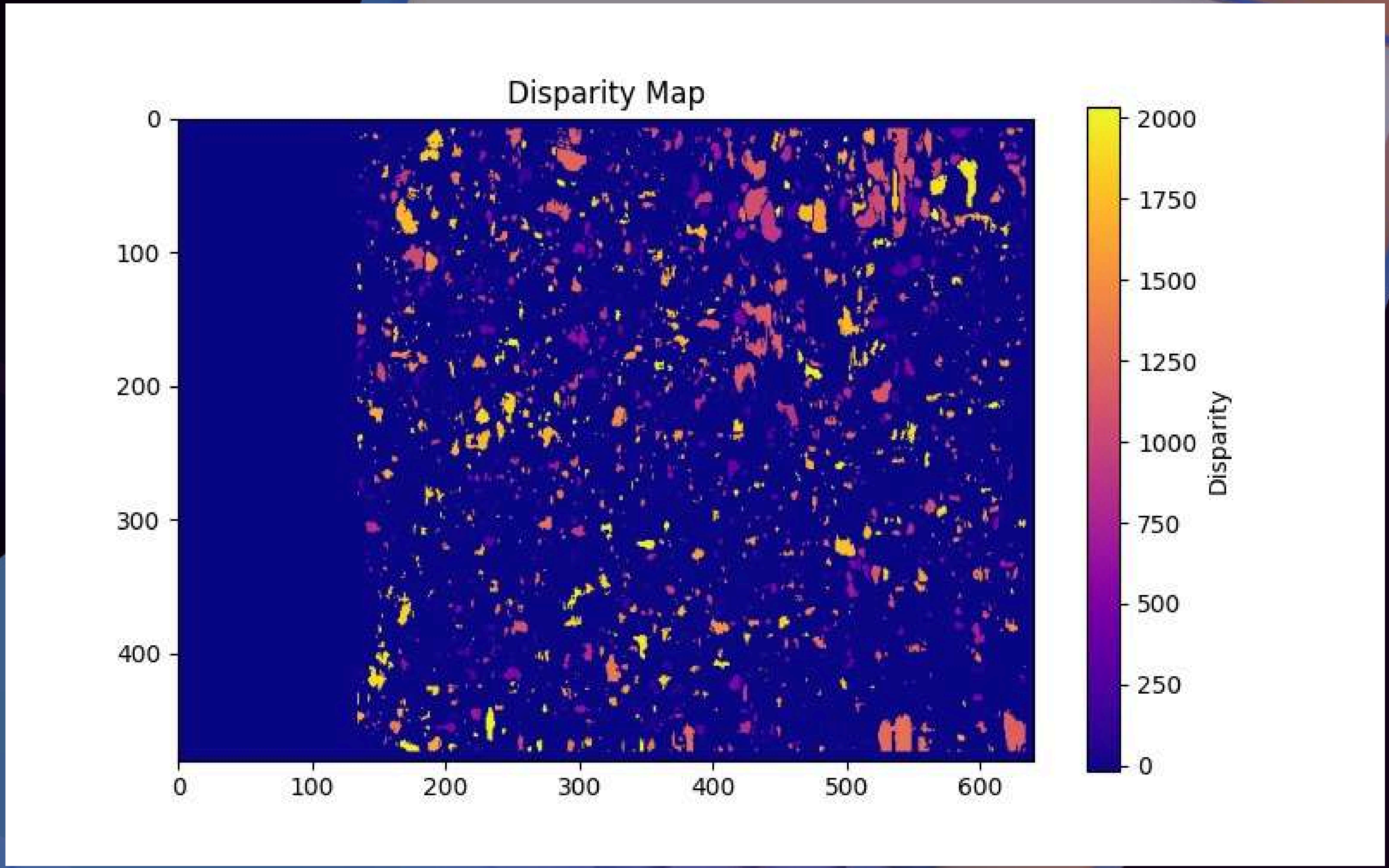
STEREO CAMERA GEOMETRY

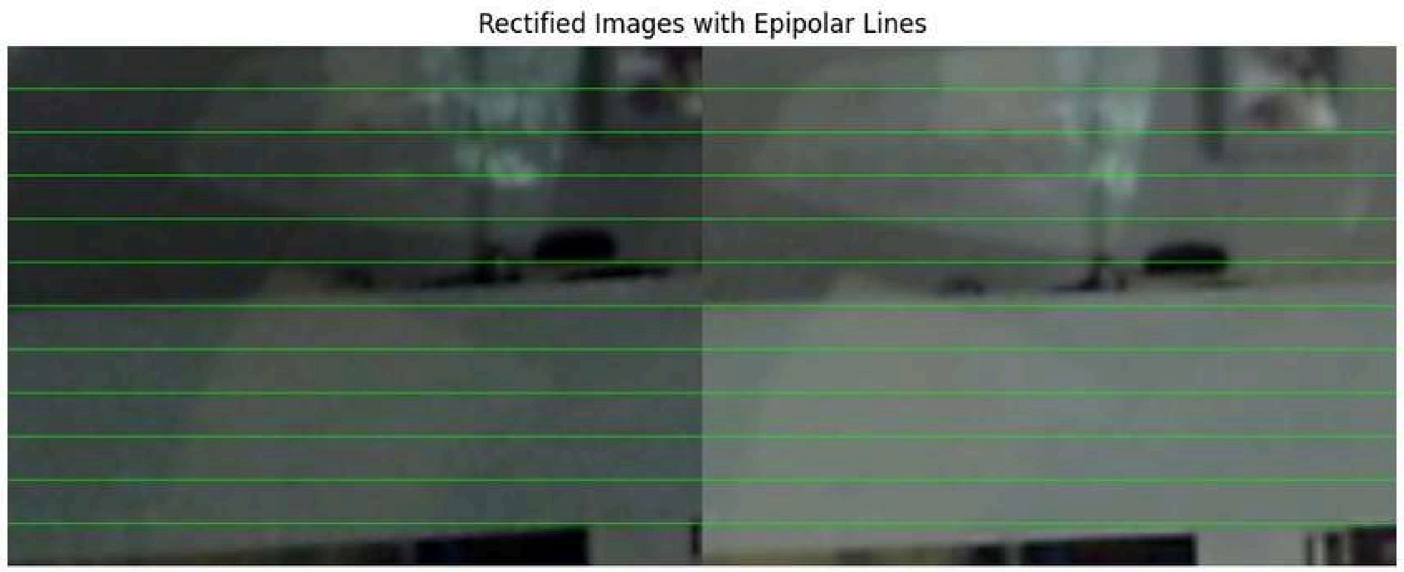


3D CLOUD MAP


3D Point Cloud
1000
800
600
400
200
0
Z
60
40
20
0
−20
Y
−20
−10
0
10
20
30
40
X

DISPARITY MAP





RECTIFIED

The rectified image shows the left and right camera views aligned so that corresponding points lie on the same horizontal lines, which is confirmed by the straight, parallel horizontal epipolar lines visible across both images. This indicates that the stereo rectification was successful.

STEREO DEPTH
(USING VIDEO TABLET)

STEREO DEPTH
(USING VIDEO TABLET)

CONCLUSION

- Successfully integrated YOLOv8n and stereo vision for vehicle detection and distance estimation
- YOLOv8n enables real-time detection of cars and trucks with a lightweight model
- Stereo vision provides depth and distance information based on image disparity
- Calibration and rectification results show reliable stereo geometry with low reprojection error
- The system delivers spatial awareness, not only object detection